Roll No. | | | | | | | | | | | | | |

Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy(Sem.-5)

MEDICINAL CHEMISTRY II-THEORY

Subject Code : BP501T

M.Code : 76786

Date of Examination : 21-11-2025

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carryingTEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. Write briefly:

   a) Why does Cetirizine cause fewer CNS side effects compared to Diphenhydramine?

   b) Which proton pump inhibitor is considered the most acid-stable?

   c) Differentiate in one line: Alkylating agents *vs.* Antimetabolites.

   d) Write one difference between Isosorbide dinitrate and Nitroglycerin.

   e) Write structural difference between Amlodipine and Nifedipine.

   f) Give structure of one loop diuretic and one thiazide diuretic.

   g) Write structure and therapeutic use of Quinidine sulphate.

   h) Mention one therapeutic use of Clofibrate.

   i) Give structures of any two oral contraceptives.

   j) Describe physiological importance of Thyroid hormones.

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SECTION - B

2. Describe the characteristic features of different classes of steroidal hormones, citing one naturally occurring compound for each class.

3. a) Classify anti-diabetic drugs with special emphasis on thiazolidinediones (Glitazones).

   b) Give the synthesis and mechanism of action of Glitazones.

4. a) Classify local anaesthetics on the basis of their basic chemical structure.

   b) Explain the structure-activity relationship (SAR) of local anaesthetics.

SECTION-C

5. Discuss the synthesis and pharmacological importance of Diphenhydraminehydrochloride.

6. Write a short note on anthracycline antibiotics.

7. Explain the mechanism of action and clinical uses of Enalapril and Lisinopril.

8. Give synthesis and mechanism of action of Methyldopa.

9. Explain the differences between class IB and IC anti-arrhythmic drugs.

10. Explain the stereochemical importance of the A/B ring junction in steroids and its effect on biological activity.

11. Discuss the advantages and disadvantages of combined oral contraceptives.

12. Explain the therapeutic use of L-thyroxine and L-triiodothyronine in hypothyroidism, including differences in onset and duration of action.

13. Discuss the structure of insulin.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

2025